

```
import pandas as pd

import numpy as np

import scipy.stats as stats

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import LabelEncoder


# -----

# Load Dataset

# -----

df = pd.read_excel("/content/student_marks_dataset.csv")


# -----

# Basic Data Inspection

# -----

print(df.head())

print(df.shape)

print(df.columns)

print(df.dtypes)


# -----

# Check for Missing Values

# -----

print(

"\nMissing Values:\n", df.isnull().sum())


# -----

# Display Rows with Missing Values

# -----

# Using 'Math' column for consistency, although it has no missing values

print("\nRows with missing 'Math' score:\n", df[df['Math'].isnull()])
```

```

# -----
# Handle Missing Values
# -----

# Using MEAN for numeric continuous data
df['Physics'] = df['Physics'].fillna(df['Physics'].mean())
df['Chemistry'] = df['Chemistry'].fillna(df['Chemistry'].mean())
df['English'] = df['English'].fillna(df['English'].mean())
df['Computer'] = df['Computer'].fillna(df['Computer'].mean())

# -----

# Label Encoding (Categorical) - Removed as 'gender' column is not present
# If needed for 'Result' column, it would be:
# le = LabelEncoder()
# df['Result'] = le.fit_transform(df['Result'])

# -----

# Drop remaining missing values (if any) - not needed after imputation
# -----

# df = df.dropna()

# -----

# OUTLIER DETECTION
# -----

# Boxplot visualization
col = ['Math', 'English', 'Computer']
df.boxplot(col)
plt.title("Boxplot for Outlier Detection")
plt.show()

```

```

# Z-score method

z = np.abs(stats.zscore(df[col]))

threshold = 3

outliers_z = np.where(z > threshold)

print("\nOutliers using Z-score:\n", outliers_z)


# IQR Method (for 'English' score)

Q1 = df['English'].quantile(0.25)

Q3 = df['English'].quantile(0.75)

IQR = Q3 - Q1


lwr_bound = Q1 - (1.5 * IQR)

upr_bound = Q3 + (1.5 * IQR)


print("\nLower Bound:", lwr_bound)

print("Upper Bound:", upr_bound)


# Identify outliers

r_outliers = df[(df['English'] < lwr_bound) | (df['English'] > upr_bound)]

print("\nIQR Outliers:\n", r_outliers)


# -----

# HANDLE OUTLIERS

# -----


# Capping (Reason: keeps dataset size same)

df['English'] = np.where(df['English'] < lwr_bound, lwr_bound, df['English'])

df['English'] = np.where(df['English'] > upr_bound, upr_bound, df['English'])


# -----

# Scatter Plot

```

```

# -----

plt.figure(figsize=(8,5))
plt.scatter(df['Math'], df['English'])
plt.xlabel("Math Score")
plt.ylabel("English Score")
plt.title("Scatter Plot")
plt.show()

# -----

# DATA TRANSFORMATION

# -----

# Log transformation (Reason: reduces skewness, makes distribution normal-like)
df['log_math'] = np.log10(df['Math'] + 1)

# Before transformation
plt.figure()
df['Math'].hist()
plt.title("Original Math Score Distribution")
plt.show()

# After transformation
plt.figure()
df['log_math'].hist()
plt.title("Log Transformed Math Score")
plt.show()

# -----

# Final Dataset

# -----

print("\nFinal Cleaned Dataset:\n", df.head())

```

